CHAPTER 02

Isomers

Independent exercise solving

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

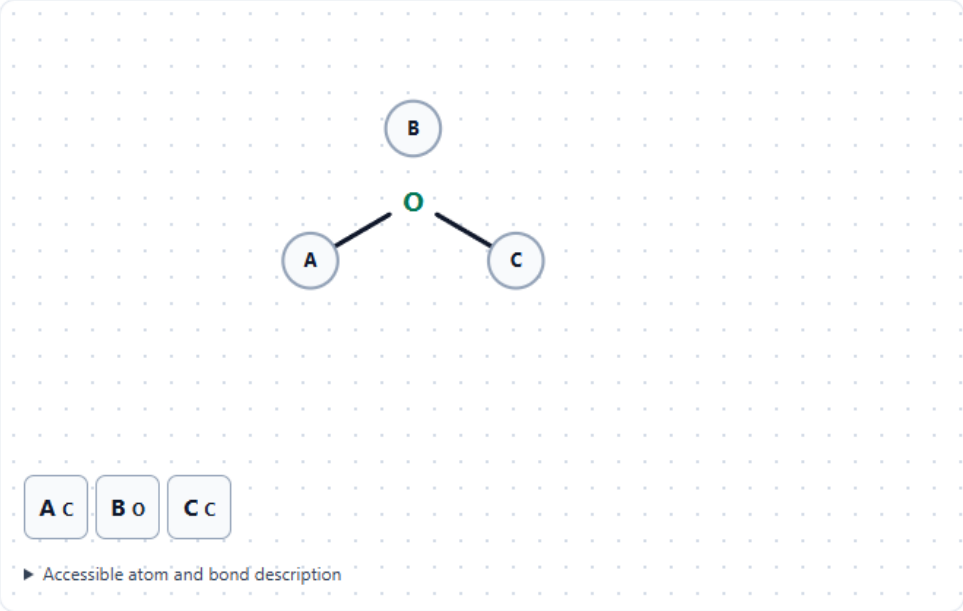
Source snapshot SHA-256: 9c4ff63c4f092cf19afe6bb2fedefb5b79ab3198072552c72d49a826d540ad22

BLIND PASS: answer keys, author molecule names/formulas, hints, explanations and answer overlays are withheld. Solve every numbered stage independently. Report the stage ID, answer, derivation, and uncertainty. Question wording and displayed diagram captions are retained because they are visible task data.

Do not open the full review packet in this chat before saving your independent attempt. The full packet contains keys and lessons that would contaminate a blind attempt.

isomer-pair | Independent exercise

Screenshot: isomer-pair-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=60.00, y=110.00; charge=0; explicit H override=none

1 (B): O; x=120.00, y=75.00; charge=0; explicit H override=none

2 (C): C; x=180.00, y=110.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1]]

isomer-pair:stage-1 | Select the oxygen atom.

Answer type: atoms

isomer-pair:stage-2 | Compared with ethanol (C2H6O), this compound is a…

Answer type: choice

Option 0 (A): Constitutional isomer

Option 1 (B): Conformer

Option 2 (C): Enantiomer

isomer-pair:stage-3 | Which can donate an O–H hydrogen bond?

Answer type: choice

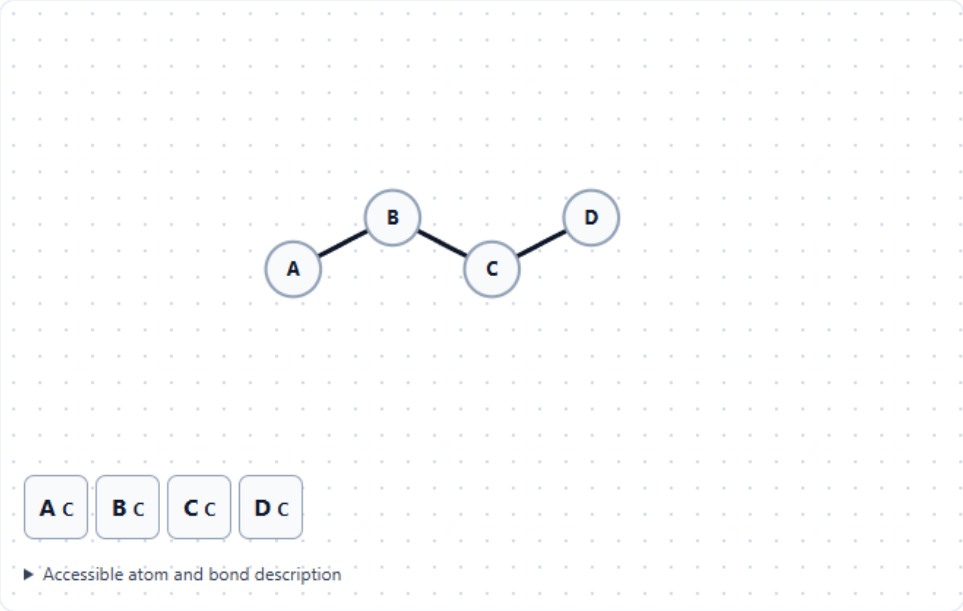
Option 0 (A): Both

Option 1 (B): Ethanol only

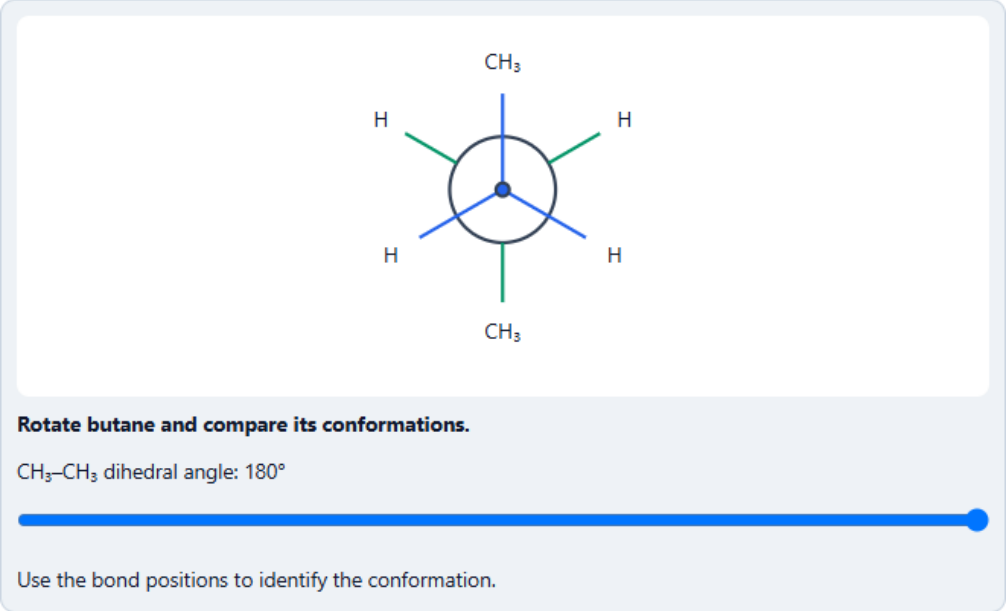
Option 2 (C): This ether only

butane-rotation | Independent exercise

Screenshot: butane-rotation-stage-1-molecule



Screenshot: butane-rotation-diagram



{"diagramSpecification": {"type": "newman", "caption": "Rotate butane and compare its conformations."}}

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1]]

butane-rotation:stage-1 | Select the two carbons forming the central bond.

Answer type: atoms

butane-rotation:stage-2 | At a 180° methyl–methyl dihedral, the conformation is…

Answer type: choice

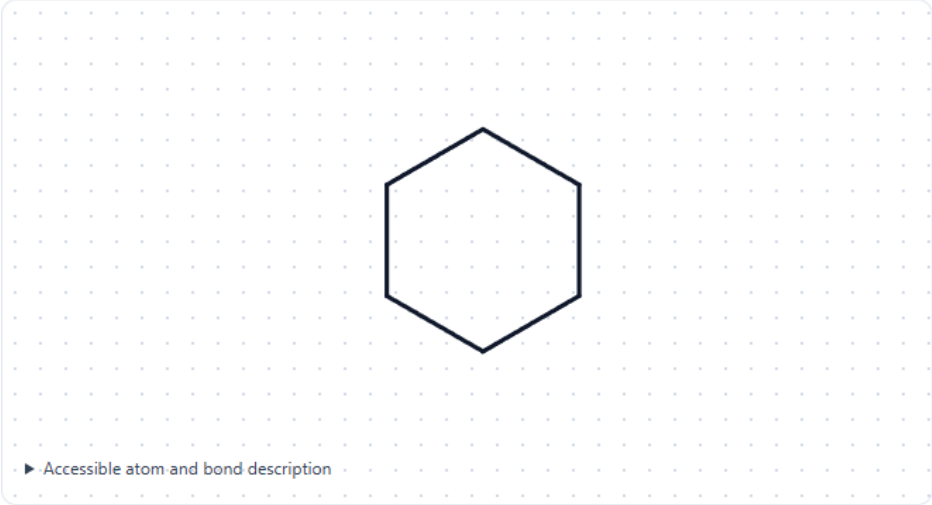
Option 0 (A): Anti

Option 1 (B): Gauche

Option 2 (C): Totally (fully) eclipsed

This stage uses a different molecule (connectivity shown).

Screenshot: butane-rotation-stage-3-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=170.00, y=37.00; charge=0; explicit H override=none

1 (B): C; x=228.89, y=71.00; charge=0; explicit H override=none

2 (C): C; x=228.89, y=139.00; charge=0; explicit H override=none

3 (D): C; x=170.00, y=173.00; charge=0; explicit H override=none

4 (E): C; x=111.11, y=139.00; charge=0; explicit H override=none

5 (F): C; x=111.11, y=71.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 5, 1], [5, 0, 1]]

butane-rotation:stage-3 | For cyclohexane (connectivity shown), a chair flip changes…

Answer type: choice

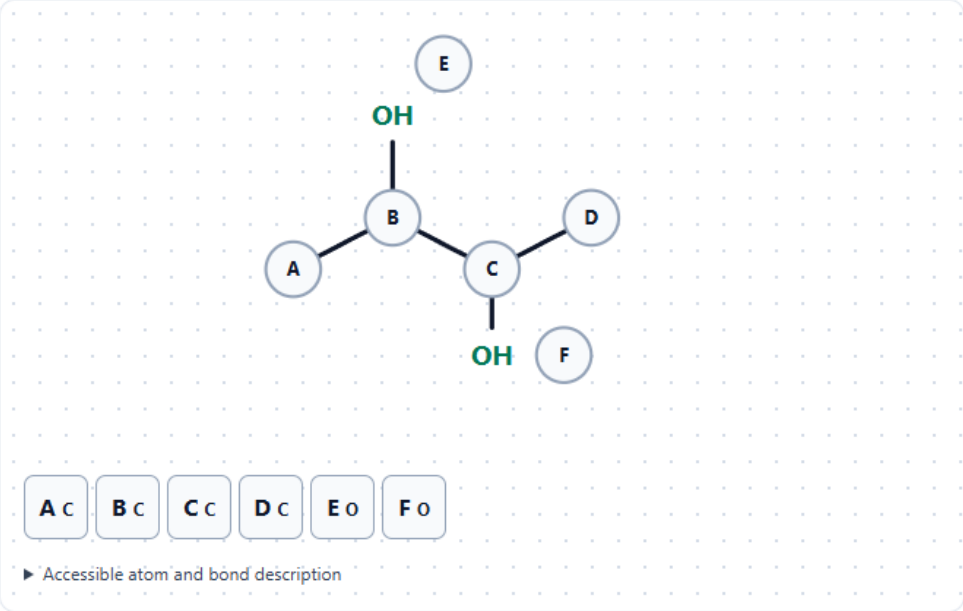
Option 0 (A): Up into down

Option 1 (B): Axial into equatorial, preserving up/down

Option 2 (C): Cis into trans

stereocenters | Independent exercise

Screenshot: stereocenters-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=25.00; charge=0; explicit H override=none

5 (F): O; x=166.00, y=165.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [1, 4, 1], [2, 5, 1]]

stereocenters:stage-1 | Select both potentially stereogenic carbons.

Answer type: atoms

stereocenters:stage-2 | What is the maximum 2^n count before symmetry is considered?

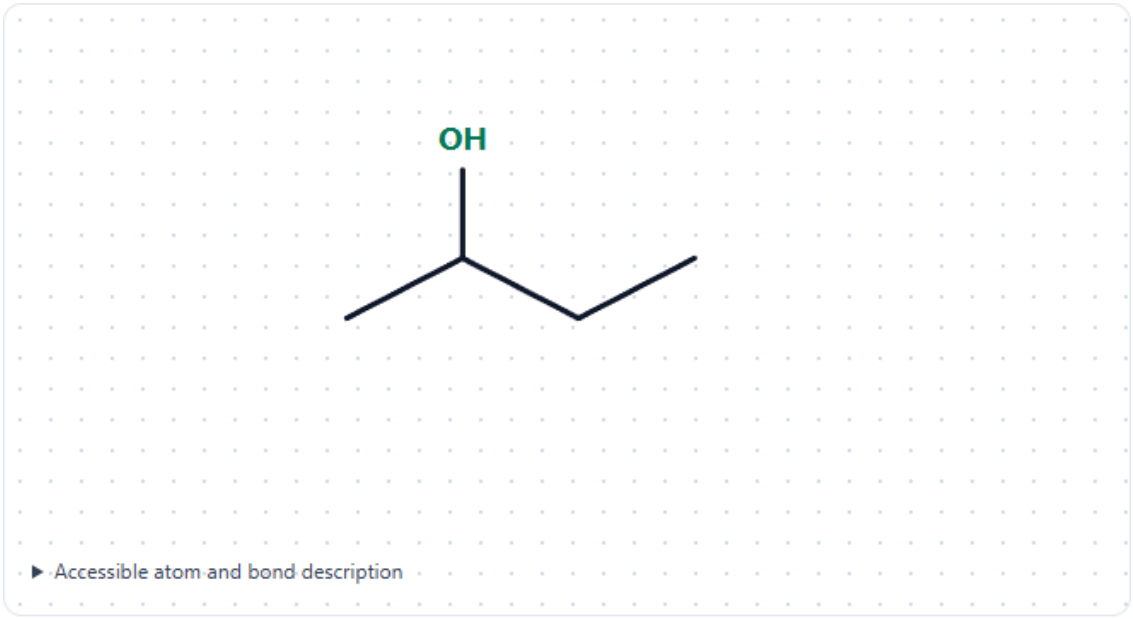
Answer type: number | unit: isomers

stereocenters:stage-3 | After accounting for the meso form, how many distinct stereoisomers exist?

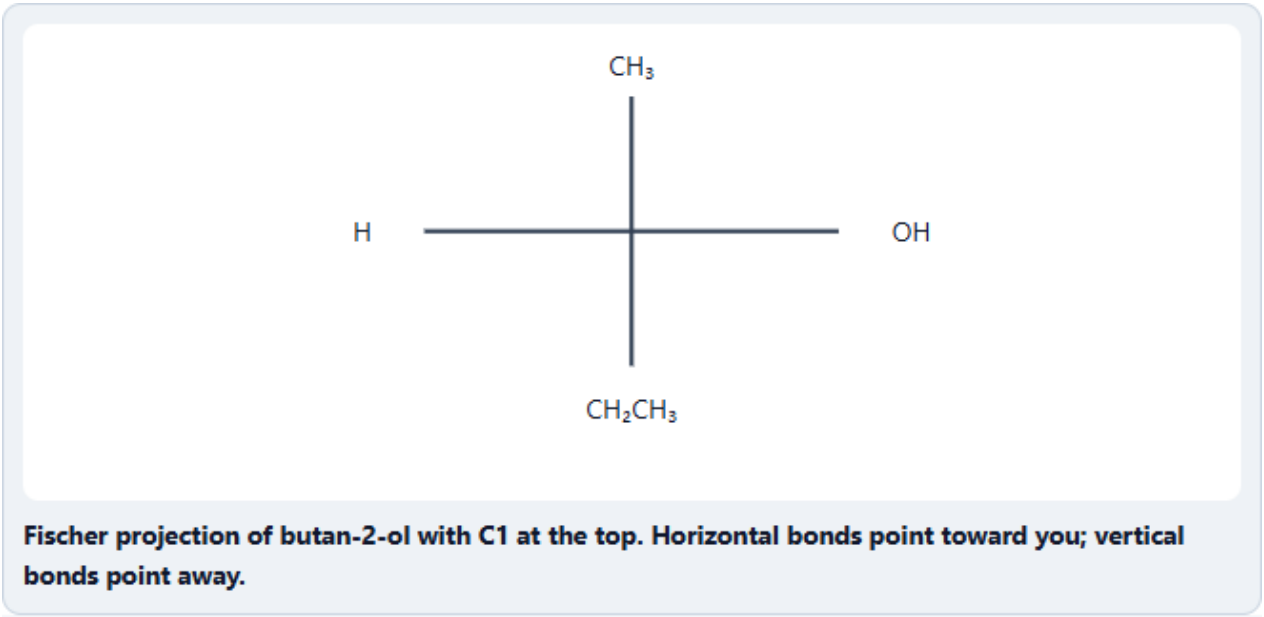
Answer type: number | unit: isomers

fischer-cip | Independent exercise

Screenshot: fischer-cip-stage-1-molecule



Screenshot: fischer-cip-diagram



Fischer projection of butan-2-ol with C1 at the top. Horizontal bonds point toward you; vertical bonds point away.

{"diagramSpecification": {"type": "fischer", "top": "CH_3", "bottom": "CH_2CH_3", "left": "H", "right": "OH", "caption": "Fischer projection of butan-2-ol with C1 at the top. Horizontal bonds point toward you; vertical bonds point away."}}

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [1, 4, 1]]

fischer-cip:stage-1 | Which substituent has highest CIP priority?

Answer type: choice

Option 0 (A): OH

Option 1 (B): Ethyl

Option 2 (C): Methyl

Option 3 (D): H

fischer-cip:stage-2 | In this projection, the lowest-priority H points…

Answer type: choice

Option 0 (A): Away

Option 1 (B): Toward you

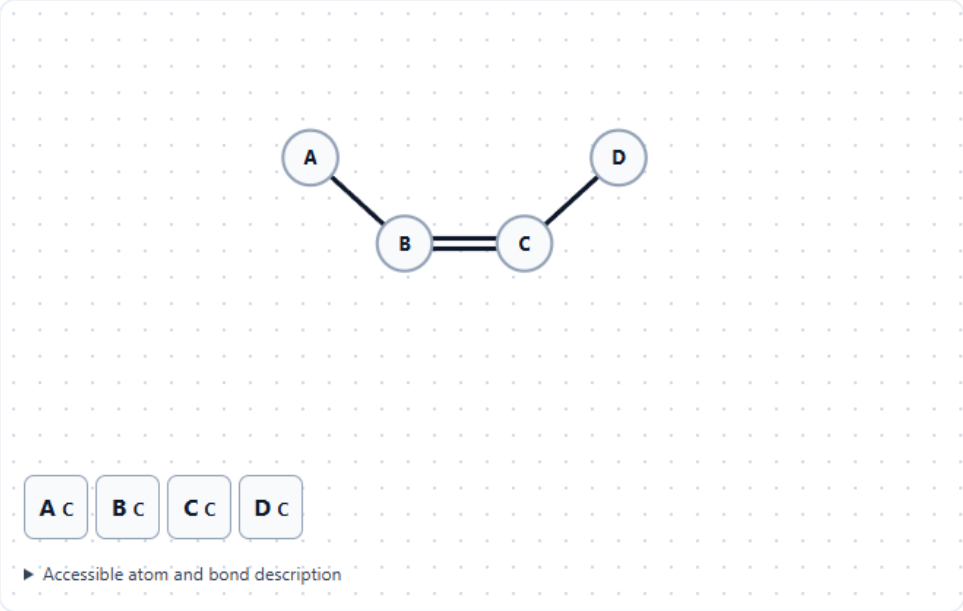
Option 2 (C): In the paper plane

fischer-cip:stage-3 | Assign the configuration: R or S.

Answer type: text

ez-chirality | Independent exercise

Screenshot: ez-chirality-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=60.00, y=50.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=100.00; charge=0; explicit H override=none

2 (C): C; x=185.00, y=100.00; charge=0; explicit H override=none

3 (D): C; x=240.00, y=50.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [2, 3, 1]]

ez-chirality:stage-1 | Select both alkene carbon atoms.

Answer type: atoms

ez-chirality:stage-2 | Assign E or Z.

Answer type: text

ez-chirality:stage-3 | Why does a pure sample of this alkene isomer show no ordinary optical rotation?

Answer type: choice

Option 0 (A): It is achiral

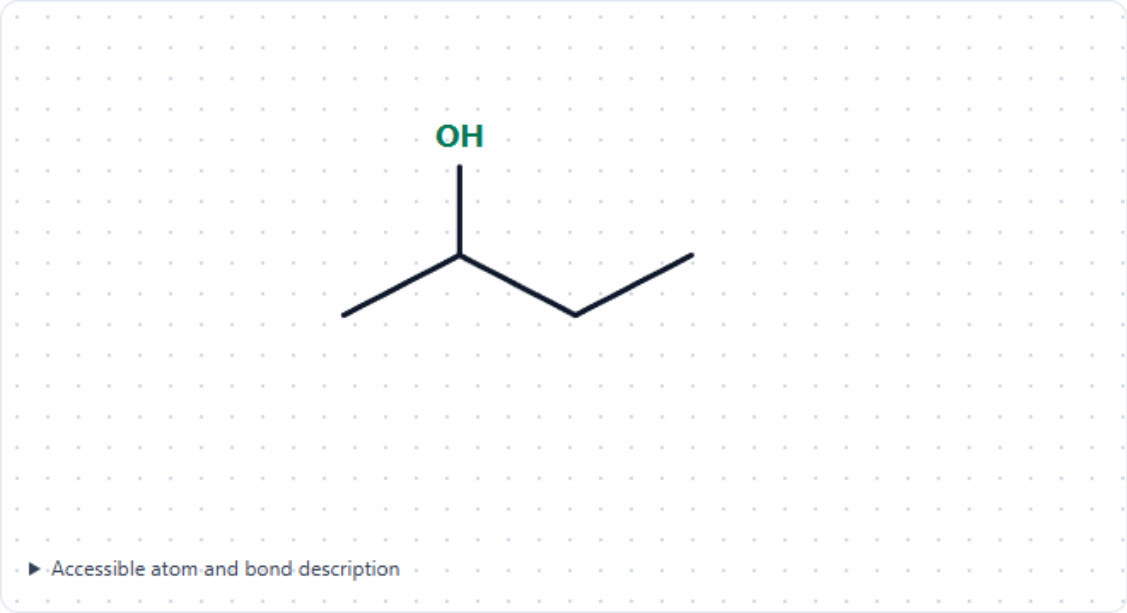
Option 1 (B): It is necessarily a 50:50 enantiomer mixture

Option 2 (C): Every Z alkene is chiral

specific-rotation | Independent exercise

Hypothetical polarimetry data for an enantiomerically pure chiral sample with the illustrated connectivity: αobs = +2.4°, c = 0.080 g/mL, path length = 5.0 cm. These are practice data, not literature measurements for butan-2-ol. Do not infer the sign from the flat molecular drawing.

Screenshot: specific-rotation-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [1, 4, 1]]

specific-rotation:stage-1 | Convert the 5.0 cm sample path length to dm.

Answer type: number | unit: dm

specific-rotation:stage-2 | For αobs = +2.4°, c = 0.080 g/mL and l = 0.50 dm, calculate [α].

Answer type: number | unit: °·mL·g^{-1}·dm^{-1}

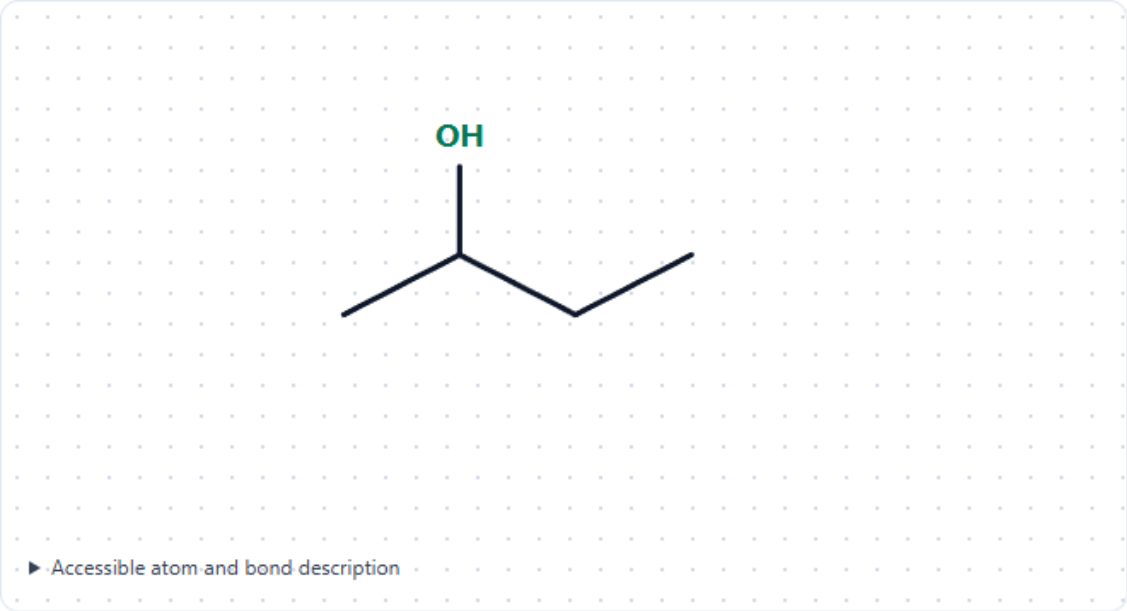
specific-rotation:stage-3 | At the same wavelength, temperature and solvent, predict αobs when c = 0.040 g/mL and l = 2.0 dm.

Answer type: number | unit: degrees

optical-mixture | Independent exercise

The illustrated chiral connectivity represents a pair of enantiomers. The 85:15 composition and +40° reference rotation are hypothetical practice data; no (+)/(−) assignment is made from the drawing.

Screenshot: optical-mixture-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [1, 4, 1]]

optical-mixture:stage-1 | The major enantiomer is dextrorotatory. Which sign describes its observed rotation?

Answer type: choice

Option 0 (A): Positive (+)

Option 1 (B): Negative (−)

Option 2 (C): The sign follows R/S without measurement

optical-mixture:stage-2 | A mixture contains 85% of that enantiomer and 15% of its mirror image. What is ee?

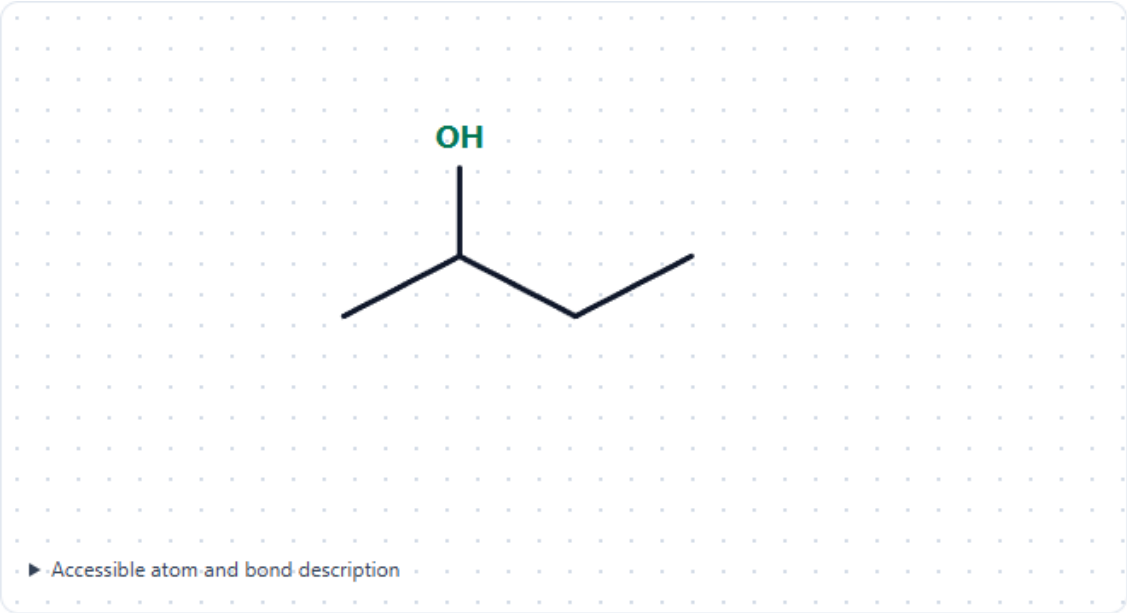
Answer type: number | unit: %

optical-mixture:stage-3 | The pure major enantiomer gives +40° at the same total concentration, path length and other conditions. Predict the mixture rotation.

Answer type: number | unit: degrees

chiral-resolution | Independent exercise

Screenshot: chiral-resolution-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [1, 4, 1]]

chiral-resolution:stage-1 | Why does ordinary achiral distillation generally fail to resolve two enantiomers?

Answer type: choice

Option 0 (A): They have the same boiling point under achiral conditions

Option 1 (B): They always have different molecular formulas

Option 2 (C): They are constitutional isomers

chiral-resolution:stage-2 | For a suitable substrate, reaction with one enantiomer of a resolving agent can create…

Answer type: choice

Option 0 (A): Separable diastereomeric derivatives or salts

Option 1 (B): Two identical achiral molecules in every case

Option 2 (C): A change in molecular formula that proves R configuration

chiral-resolution:stage-3 | A remote substituent change retains the local arrangement at a tetrahedral stereocenter but changes its CIP priorities. Its R/S label…

Answer type: choice

Option 0 (A): May change without physical inversion

Option 1 (B): Must stay the same whenever geometry is retained

Option 2 (C): Determines whether optical rotation is positive